Additional file 1. Comparator implementation and its validation

This file supports Section 6.1 of the main text and Section 3.4 of the main text. It states what the correlation-module comparator used here shares with the canonical WGCNA pipeline, where it departs, and whether any conclusion in the manuscript depends on those departures.

Why the comparators are implemented directly

The benchmark repeats every selection strategy across 26 cohort subsets and every leave-one-dataset-out fold within each subset. Calling reference packages across language boundaries at that repetition count is slow and, more importantly, makes the result depend on the versions of several external runtimes. Implementing the comparators in a single environment keeps the whole benchmark reproducible from one command. The cost is that these are not the reference implementations, and we do not claim they reproduce them.

What the hub ranking shares with WGCNA, and where it departs

Element	Canonical WGCNA	Used here	Status
Similarity measure	Pearson correlation	Pearson correlation	same
Adjacency	signed or unsigned power	unsigned power	restricted
Soft-threshold power	chosen by scale-free fit	fixed at $\beta = 6$	differs
Module detection	TOM distance, dynamic tree cut	clustering on correlation distance	differs
Hub score	kME, eigengene correlation	intramodular connectivity kIN	differs
Module eigengene	first principal component	not computed	absent

The departures are real and are why the manuscript calls this a correlation-module comparator rather than WGCNA. The question the rest of this file answers is whether any of them changes a conclusion.

Scale-free topology fit

The canonical criterion selects the smallest power whose scale-free fit R^2 reaches about 0.80. The fit across powers on the harmonised discovery data is below.

power	scale free r2
2.000	0.198
4.000	0.283
6.000	0.619
8.000	0.762
10.000	0.862

power	scale free r2
12.000	0.871
14.000	0.897

At the power used here ($\beta = 6$) the fit is $R^2 = 0.62$, below the canonical threshold; the best fit over the grid is 0.90 at $\beta = 14$. A lower power yields a denser network and therefore larger modules. Section 3.4 of the main text shows that larger modules favour the procurement-tracking module, so this departure works against the manuscript argument rather than for it.

The variant grid

Every combination of the departing choices was run: 32 variants over module definition, ranking statistic, adjacency and soft-threshold power. The question is whether any combination places immediate-early genes outside the top of the ranking, which would mean the procurement signal is an artefact of our implementation rather than of the data. **0 of 32 variants place an immediate-early gene in the top 50.** The result does not depend on the implementation choices.

definition	rank by	adjacency	power	module size	IEG in module	IEG in top 50	IEG median rank
top-decile |gs|	kIN	unsigned	4	990	12	0	958
top-decile |gs|	kME	unsigned	4	990	12	0	966
top-decile |gs|	kIN	signed	4	990	12	0	969
top-decile |gs|	kME	signed	4	990	12	0	966
top-decile |gs|	kIN	unsigned	6	990	12	0	919
top-decile |gs|	kME	unsigned	6	990	12	0	966
top-decile |gs|	kIN	signed	6	990	12	0	968
top-decile |gs|	kME	signed	6	990	12	0	966
top-decile |gs|	kIN	unsigned	8	990	12	0	838
top-decile |gs|	kME	unsigned	8	990	12	0	966
top-decile |gs|	kIN	signed	8	990	12	0	962
top-decile |gs|	kME	signed	8	990	12	0	966
top-decile |gs|	kIN	unsigned	12	990	12	0	664
top-decile |gs|	kME	unsigned	12	990	12	0	966
top-decile |gs|	kIN	signed	12	990	12	0	936
top-decile |gs|	kME	signed	12	990	12	0	966
TOM cluster (k=4)	kIN	unsigned	4	3956	17	0	3531
TOM cluster (k=4)	kME	unsigned	4	3956	17	0	3501
TOM cluster (k=4)	kIN	signed	4	2068	0	0	–
TOM cluster (k=4)	kME	signed	4	2068	0	0	–
TOM cluster (k=4)	kIN	unsigned	6	3957	17	0	3036
TOM cluster (k=4)	kME	unsigned	6	3957	17	0	3501

definition		rank by	adjacency	power	module size	IEG in module	IEG in top 50	IEG median rank
TOM (k=4)	cluster	kIN	signed	6	2066	0	0	–
TOM (k=4)	cluster	kME	signed	6	2066	0	0	–
TOM (k=4)	cluster	kIN	unsigned	8	3957	17	0	2579
TOM (k=4)	cluster	kME	unsigned	8	3957	17	0	3501
TOM (k=4)	cluster	kIN	signed	8	2063	0	0	–
TOM (k=4)	cluster	kME	signed	8	2063	0	0	–
TOM (k=4)	cluster	kIN	unsigned	12	3957	17	0	1710
TOM (k=4)	cluster	kME	unsigned	12	3957	17	0	3501
TOM (k=4)	cluster	kIN	signed	12	2059	0	0	–
TOM (k=4)	cluster	kME	signed	12	2059	0	0	–

Provenance of the remaining selectors

The same reasoning applies to the other three strategies in the benchmark. Each is stated here with the published definition it stands for and the respects in which the implementation used here is narrower.

Strategy	Stands for	How the implementation is narrower
DEG_meta	Per-cohort Welch t, combined by Stouffer	weights are $\sqrt{n}$; no random-effects variance component
RBS	Stability selection (Meinshausen and Bühlmann)	bootstrap selection inside each cohort, aggregated by geometric mean across cohorts, with a sign-concordance gate and a perturbation term. It is used as an instrument standing in for this family, not proposed as a contribution.
RBS_orth	The same, after residualisation	every gene is residualised on a per-sample handling score before selection
Baseline selectors inside RBS	univariate F, LASSO, elastic net, mRMR, ReliefF, random forest, Boruta	all implemented directly rather than through packages that pin older numpy or scikit-learn versions

None of these is offered as a new method. They are present so that the benchmark contains a representative of each family a reader would expect to see compared.

Block-size simulation with the RBS implementation

Section 3.4 of the main text reports that the connectivity criterion picks up more of the confounder than the stability criterion when the confounder block is large and less when it is small. The table gives the paired difference in confounder fraction of the top 50, connectivity minus RBS, over 25 replicates per configuration, run with the actual RBS implementation rather than a stand-in.

Disease	Confounder	Mean	CI low	CI high	Wilcoxon p	Reps	Connectivity
20	200	+0.282	+0.236	+0.328	1.2e-05	25	worse
30	30	+0.064	+0.038	+0.090	4.4e-04	25	worse
100	30	-0.289	-0.356	-0.221	1.8e-05	25	better
100	100	+0.128	+0.022	+0.234	3.0e-02	25	worse
200	20	-0.213	-0.268	-0.158	8.5e-05	25	better

Disease and Confounder are the block sizes; Mean and the 95% CI are the paired difference; Connectivity states whether the connectivity criterion did worse or better. Positive values mean the connectivity criterion placed more confounder genes in its top 50 than the stability criterion did. The sign follows the block-size ratio, which is the mechanism described in Section 3.4 of the main text.